IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Tuesday, 5 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : E. Kerrigan

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1. Consider the feedback system in Figure 1. Here, $G(s)$ is a nominal system model and $K(s)$ is a controller. The transfer matrices $\Delta_1(s)$ and $\Delta_2(s)$ denote stable uncertainties.

The robust stability specification is to design a controller $K(s)$ such that the feedback loop is internally stable for all $\Delta_1(s)$ and $\Delta_2(s)$ satisfying $\|\Delta(j\omega)\| < 1$ for all real ω where

$$\Delta(s) = \begin{bmatrix} \Delta_1(s) \\ \Delta_2(s) \end{bmatrix}.$$

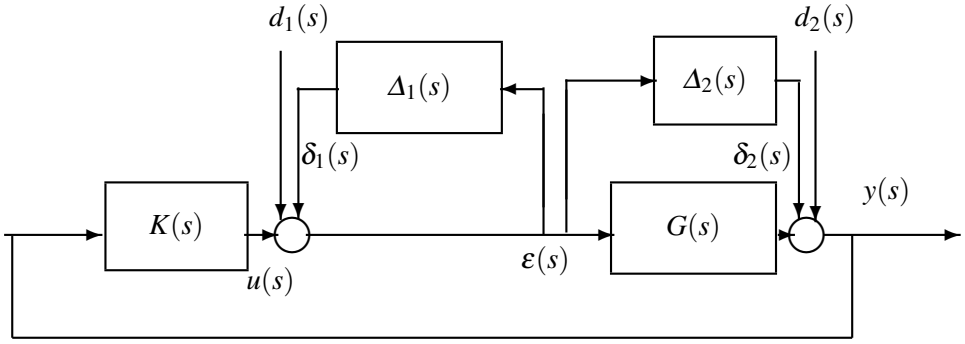


Figure 1

a) State the small gain theorem concerning the internal stability of a feedback loop with a forward transfer matrix Δ and a feedback transfer matrix H. [5]

b) Give an interpretation of the small gain theorem for single-input-single-output systems in terms of the Nyquist stability criterion. [5]

c) Use the small gain theorem to derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$ and $K(s)$ that are sufficient to achieve the robustness specifications. [5]

d) Derive and draw a generalised regulator problem that captures these sufficient conditions. [5]

Hints:

- *You may want to express ε in terms of $\delta := \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix}$, ignoring d_1 and d_2.*
- *You may want to express ε in terms of $d := \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$, ignoring δ_1 and δ_2.*

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2. a) Consider a state–variable model described by the dynamics

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t),$$

and let $H(s) = D + C(sI - A)^{-1}B$ denote the corresponding transfer matrix.

Let $\gamma > 0$ be given. Derive sufficient conditions in the form of LMIs for the stability of A and for $\|H\|_\infty < \gamma$. [5]

b) Consider the static output feedback problem in Figure 2. Let $T_{yr}(s)$ denote the transfer matrix from r to y. An internally stabilizing static gain matrix K is to be designed such that, for given $\gamma > 0$, $\|T_{yr}\|_\infty < \gamma$.

i) Derive a state space realization for $T_{yr}(s)$. [5]

ii) Derive sufficient conditions for the existence of a feasible K. Your conditions should be in the form of the existence of solutions to matrix inequalities, which are not necessarily linear, involving a Lyapunov variable P. [5]

iii) Consider the following approach for linearizing the matrix inequality derived above. Introduce a variable X that satisfies the linear equation $PB = BX$. Use this to derive sufficient conditions for the existence of a feasible K. State a further assumption on X that guarantees the existence of K. [5]

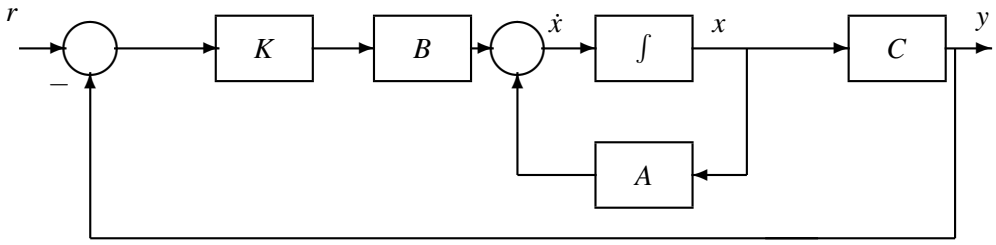


Figure 2

3. Consider the linear quadratic regulator (LQR) in Figure 3 for which $x(0) = x_0$, where $d(t)$ denotes a disturbance. Let $z = \begin{bmatrix} z_1^T & z_2^T \end{bmatrix}^T$. A stabilizing state-feedback gain matrix F is to be designed such that, when $d(t) = 0, \forall t$, the cost function $J = \|z\|_2^2$ is minimized.

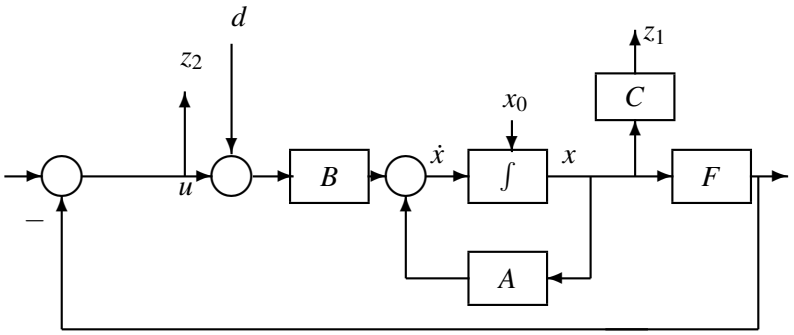


Figure 3

a) Define a suitable Lyapunov function $V(t)$ and obtain an expression for $\int_0^\infty \dot{V}(t)dt$. State clearly any assumptions made. [5]

b) Find an expression for F that minimizes J. Give also the algebraic Riccati equation that needs to be satisfied. What is the minimum value of J? [5]

c) Prove now that the closed-loop system in Figure 3 is stable. State the assumptions required to guarantee stability. [5]

d) By evaluating a state–space realisation for $T_{z_1 d}(s)$ in Figure 3 and using the Bounded Real Lemma, derive an upper bound on $\|T_{z_1 d}\|_\infty$. Ignore issues relating to strict matrix inequalities. [5]

4. Consider the $\mathscr{H}_\infty$ regulator given by

$$\begin{bmatrix} z(s) \\ y(s) \end{bmatrix} = P(s) \begin{bmatrix} w(s) \\ u(s) \end{bmatrix},$$
$$u(s) = Fy(s),$$

where

$$P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \stackrel{s}{=} \left[\begin{array}{c|c|c} A & B_1 & B_2 \\ \hline C & 0 & 0 \\ 0 & 0 & I \\ \hline I & 0 & 0 \end{array}\right],$$

for which it is assumed that the triples (A,B_1,C) and (A,B_2,C) are minimal and the state of the system is initially at rest. Let $H(s)$ denote the transfer matrix from w to $z = [\, z_1^T \;\; z_2^T \,]^T$ and let $\gamma > 0$ be given.

A stabilizing state-feedback gain matrix F is to be designed such that $\|H\|_\infty \leq \gamma$.

a) Write down a block diagram representing the feedback loop for the generalised regulator. Clearly indicate all the signals. [5]

b) By defining a suitable Lyapunov function and a cost function J involving an auxiliary variable $X = X'$, and carrying out two completions of squares, derive an expression for J that can be used to solve the design problem. [5]

c) Use the expression for J to formulate the design problem and derive sufficient conditions for its solution. Your conditions should be in the form of the existence of a certain solution to an algebraic Riccati equation. It should also include an expression for F and an expression for the worst–case disturbance w^*. [5]

d) Suppose that $B_1 = B_2 =: B$ and let $\gamma = 1$. Write down the Riccati equation and give a condition for the existence of a stabilizing solution. [5]

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